

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q19-1196

Speed 138 kV
January 28, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1196. The Interconnection Customer proposes a 187.9 MW wind facility interconnecting at the Speed 138 kV substation in the Kentucky area. The request was received on April 28, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 345 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$1,061,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Stelco – Murphys 138 kV	Pine Bluff Watson Chapel – Carraizo 1201 345 kV	114.9	207	14.9
Hubbardstown – Tristate 345 kV	International Paper – Hanging Rock 500 kV	103.9	1,562	9.1

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Stelco – Murphys 138 kV	Rebuild line with higher-capacity conductor	\$5,432,000	16% (\$866,000)
Hubbardstown – Tristate 345 kV Line	Reconductor line	\$12,377,000	2% (\$195,000)
	Total allocated to project		\$1,061,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.